

Bojan Derajić

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PROFILE

- Research Scientist at AUMOVIO and PhD candidate at TU Berlin specializing in learning-based motion planning and control for autonomous robots. Develops methods that combine model-based control with machine learning for local navigation and collision avoidance in unknown and dynamic environments. First-author publications at CoRL, ICRA, IROS, and IEEE RA-L; 5 patent applications, including 4 as lead inventor.

EXPERIENCE

AI Research Scientist

04/2023 – Present

AI Lab Berlin, AUMOVIO (formerly Continental Automotive)

Berlin, Germany

- Research safe motion planning methods for robot navigation in unknown environments based on local onboard sensing.
- Developed neural terminal constraints for MPC-based local planning in unknown static and dynamic environments.
- Developed observation-conditioned safety filters that adapt to current sensor readings for cluttered and crowded dynamic scenes.
- Implemented and evaluated algorithms in simulation and deployed on a ground mobile robot and quadrotors using Python/C++, ROS 2, CasADi, and PyTorch.
- Optimized real-time performance for mobile robot teleoperation (BeIntelli project).

Research and Teaching Assistant

12/2021 – 03/2023

Faculty of Electrical Engineering, University of Banja Luka

Banja Luka, Bosnia-Herz.

- Developed adaptive nonlinear MPC for quadrotor trajectory tracking; taught tutorials in control systems and artificial intelligence.

EDUCATION

PhD, Electrical Engineering and Computer Science (Robotics & AI)

2023 – Present

Technical University of Berlin

Berlin, Germany

- Thesis: “Learning Reachability-Based Safety Representations for Real-Time Robot Navigation in Unknown and Dynamic Environments”
- Supervising master’s students in AI-enhanced robotics and control.

MSc, Electrical Engineering and Computer Science (Robotics & Control)

2021 – 2022

University of Belgrade

Belgrade, Serbia

- Thesis: “Adaptive Nonlinear Model Predictive Control for Quadcopters” | GPA: 10.0/10.0.

BSc, Electrical Engineering and Computer Science (Robotics & Control)

2017 – 2021

University of Banja Luka

Banja Luka, Bosnia-Herz.

- Thesis: “Trajectory Planning for Robotic Manipulators” | GPA: 10.0/10.0.
- Awards: Golden Badge of the University of Banja Luka; Nikola Tesla Medal.

SELECTED PUBLICATIONS

- B. Derajić et al.**, “VertexCBF: Improving Neural Control Barrier Functions via Vertex-Restricted Control Search,” *CoRL*, 2026. (accepted)
- B. Derajić et al.**, “CN-CBF: Composite Neural Control Barrier Function for Robot Navigation in Dynamic Environments,” *IROS*, 2026.
- B. Derajić et al.**, “ORN-CBF: Learning Observation-conditioned Residual Neural Control Barrier Functions via Hypernetworks,” *ICRA*, 2026.
- B. Derajić et al.**, “Residual Neural Terminal Constraint for MPC-based Collision Avoidance in Dynamic Environments,” *CoRL*, 2025.
- B. Derajić et al.**, “Learning Maximal Safe Sets Using Hypernetworks for MPC-Based Local Trajectory Planning in Unknown Environments,” *IEEE RA-L*, 2025.

SKILLS

Motion Planning & Control: motion planning, nonlinear MPC, optimal control, collision avoidance, trajectory tracking, safety filtering, control barrier functions, reachability analysis, learning-based planning and control

Programming & Robotics Software: C/C++, Python, MATLAB, ROS/ROS 2, CasADi, acados, PyTorch, JAX, MLflow, Gazebo, Docker, Git, Linux